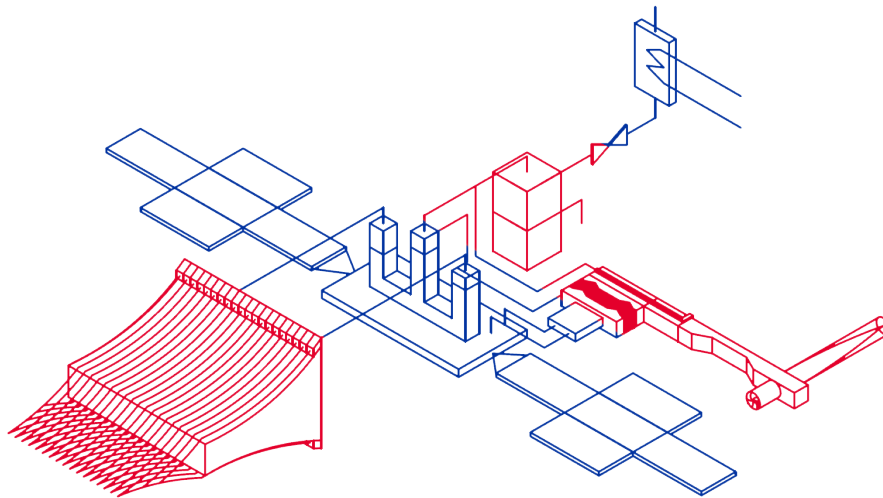


Skyraker Development Roadmap

From scratch to TRL 4.5 in 9 months

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Programme Summary

Vehicle: Ikarus – Multipurpose interplanetary spacecraft

Engine: Skyraker – ISRU-enabled hybrid propulsion subsystem

Architecture: Flash Desalination → Electrolysis → Expander-cycle Aerospike + Ion Thruster

Resources: In-situ water (Earth oceans, Mars ice, Europa, Enceladus)

University context: Nothing (this is just a suicide squad having fun)

Budget: < €500 **Target:** TRL 4.5 in 9 months

Companion reports: [1] • [2] • [3] • [4] • [5]

March 10th, 2026

Version 2.0

Part of the Space Framework™.



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1 Context and Narrative Framework

1.1 The Ikarus™ Vehicle and the Space Framework™

Skyraker is not an isolated engine. It is the **propulsion heart of Ikarus**, a multipurpose crewed spacecraft designed for interplanetary operations within the Space Framework architecture. The Space Framework™ envisions a self-sustaining cisplanetary economy based on three principles: reusability, in-situ resource utilisation (ISRU), and modular architecture.

Ikarus is designed to:

- operate in a wide range of environments (LEO, deep space, Mars orbit, icy moon flybys);
- use locally-sourced water and resources as both propellant and energy carrier;
- minimise Earth-dependent consumables through closed-loop ISRU.

Skyraker translates these objectives into hardware. Its mission within Ikarus is to provide:

1. **High-thrust chemical propulsion** (expander cycle aerospike, H_2/O_2 from on-board water electrolysis) for orbital manoeuvres, EDL, and ascent;
2. **High-Isp electric propulsion** for interplanetary cruise and RCS;
3. **Atmospheric breathing propulsion** for sustained low-altitude orbit maintenance.

Ultimate aim of this roadmap:

“Experimental feasibility study and laboratory validation of the Skyraker ISRU-enabled hybrid propulsion subsystem for the Ikarus™ spacecraft concept, within the Space Framework™ interplanetary architecture.”

1.2 Heritage and Benchmarks

Skyraker draws on, and can cite, the following flight-proven and research heritage:

System	Organisation	Relevance to Skyraker
SABRE engine	Reaction Engines Ltd	Combined-cycle airbreathing/rocket concept
ATREX	JAXA	Air-Turbo-Ramjet Expander cycle
NSTAR / Dawn	NASA/JPL	Gridded ion thruster heritage
Deep Space 1	NASA	First deep-space ion propulsion demonstration
ECLSS water recovery	NASA/ISS	ISRU water processing reference
ESA ABEP programme	ESA/SITAEL	Air-breathing electric propulsion
Aerospike (J-2S, Rocketdyne)	NASA/Rocketdyne	Aerospike nozzle heritage

Table 1: Technology heritage directly cited in Skyraker development.

2 System Architecture

The Skyraker system operates as a **closed-loop ISRU propulsion chain**:

$$\underbrace{\text{Water source}}_{\text{ocean / ice / plume}} \xrightarrow{\text{Flash distillation}} \underbrace{\text{Distilled H}_2\text{O}}_{\text{WTP output}} \xrightarrow{\text{Electrolysis}} \underbrace{\text{H}_2 + \text{O}_2}_{\text{propellant}} \xrightarrow{\text{Expander cycle}} \underbrace{\Delta v}_{\text{thrust}} \quad (1)$$

In parallel, solar energy drives flash distillation, electrolysis and the ion thruster:

$$\underbrace{\text{Solar flux}}_{\Phi_{\odot}(r)} \xrightarrow{\text{PV array}} \underbrace{P_{\text{elec}}}_{\text{DC power}} \rightarrow \begin{cases} \text{Flash distillation: pure H}_2\text{O production} \\ \text{Electrolysis: gaseous H}_2/\text{O}_2 \text{ production} \\ \text{Ion thruster: high-I}_{\text{sp}} \text{ cruise} \end{cases} \quad (2)$$

Subsystem	Description	Report
Water Treatment Plant (WTP)	Flash distillation of raw seawater/brine. Removes ionic contaminants, produces distillate for PEM feed ($C_d < 200 \mu\text{S}/\text{cm}$).	[1]
Solar-Electric Assembly (SEA)	PEM electrolysis powered by photovoltaic array. Produces H_2 and O_2 from distilled water. Faradaic efficiency $\eta_F \approx 0.90\text{--}0.95$.	[2]
eXperimental-aeroSpike Liquid Rocket (XSLR)	H_2/O_2 expander-cycle engine with aerospike nozzle, derived from JAXA's ATREX. High thrust for launch assist, orbital insertion, EDL braking, ascent in low-gravity. I_{sp} : 450 s.	[3]
Ion Cruise and Exospheric Rocket (ICER)	Microwave air plasma thruster (2.45 GHz, after [6]), for sustained low-altitude orbit maintenance at Mars, Titan, Europa etc. I_{sp} : 50–1000 s.	[4]

Table 2: Skyraker subsystem breakdown and document cross-references.

Skyraker operates in two primary propulsion modes:

Mode 1 – Chemical Using the high thrust and low I_{sp} XSLR subsystem.

Mode 2 – Electric Using the high I_{sp} and low thrust μ -wave ICER subsystem in 2 modes:

Mode 2a – Stored propellant: H_2 from electrolysis fed into the microwave cavity.

Mode 2b – Atmosphere-Breathing Electric Propulsion (ABEP): ambient atmosphere ingested directly into the ionisation chamber.

Inspiration: The multi-modal architecture draws direct inspiration from the **Ranger spacecraft** in the movie *Interstellar*: a compact vehicle capable of atmospheric entry, orbital operations, and precision docking, using an elegant propulsion system that adapts to the environment. Skyraker is the physically-grounded engineering realisation of that philosophy.

3 Development Philosophy and TRL Framework

3.1 TRL Scale Applied to Skyraker

TRL	Definition (NASA)	Skyraker equivalent
1	Basic principles observed	Physical laws governing ISRU chain identified.
2	Technology concept formulated	Skyraker concept + Space Framework document.
3	Experimental proof of concept	Individual subsystem bench tests.
4	Component validation, lab environment	Integrated ISRU chain: WTP + SEA + XSLR (steam) + ICER (μW). TARGET.
5	Component validation, relevant environment	Integrated chain powered by actual PV; XSLR on electrolytic H_2/O_2 ; ICER in low-pressure environment. Requires dedicated gas-handling facility and vacuum chamber. <i>Partially obtained as TRL 4.5.</i>
6	System/subsystem model demonstration	Flight-like hardware; full-scale vacuum test.
7	System prototype in operational environment	Suborbital test.

Table 3: TRL scale with Skyraker mapping. **TRL 4.5** is this work's 9-month target.

TRL 4 demands laboratory validation of *integrated components* in a controlled bench environment, not a complete flight system. For Skyraker, this means demonstrating:

raw water → distillation → electrolysis → gas feed → steam thrust + plasma thrust

It does **not** require at **TRL 4**:

- LH₂/LOX high-energy combustion or cryogenics;
- an ion thruster operating in vacuum (in-air demo after [6] suffices for **TRL 4**);
- PV-powered operation (lab power supply simulates PV output);
- mission simulation or trajectory analysis;
- flight-qualified hardware or space environment testing.

These advance the technology to the natural *future work* section developments.

3.2 Technology Readiness Level 4.5

Technology Readiness Level (TRL) 4.5 is an informal intermediate stage sometimes used in NASA technology development reports to indicate progress between TRL 4 and TRL 5. While the official NASA scale defines discrete levels from TRL 1 to TRL 9 [9], fractional levels such as TRL 4.5 may be used in project documentation to represent technologies that have advanced beyond laboratory breadboard validation but have not yet achieved full validation in a relevant environment.

In this project, TRL 4.5 refers to the integrated testing of all subsystems together with the controlled combustion of electrolytically produced gases. These tests will be performed in a controlled laboratory environment, as described in the following sections. Since the materials and experimental setup used to achieve this additional milestone are the same as those employed for the TRL 4 demonstration, further development will still be required to achieve full TRL 5 [10].

3.3 Budget and Resources

Phase	Estimated cost	Cumulative
Phase 0 – System Definition and Modelling	€0	€0
Phase 1 – WTP (flash desalination)	€0–€50	≤€50
Phase 2 – SEA (water Electrolysis)	€0–€50	≤€150
Phase 3A – XSLR cold flow (nozzle)	€0–€50	≤€200
Phase 3B – XSLR steam thruster	€0–€50	~€250
Phase 4 – Integration (plumbing etc.)	€0–€50	~€300
Phase 5 – ICER μ -wave-Ion thruster & Re-integration	€0–€50	<€350
Total (Phases 0–4)	~€0–€400	
Total (all phases)	<€500	

Table 4: Estimated budget by phase. Comfortably within reasonable constraints.

4 Phase 0 – System Definition and Modelling

4.1 Objectives

Phase 0 establishes the numerical baseline and the system architecture before any laboratory activity. Every subsequent phase is bounded by the requirements derived here.

4.2 Activities

4.2.1 Top-down requirements derivation

Map full-scale Skyraker parameters to laboratory-demonstrator parameters. Define scale factors for: mass flow rate, thrust, electrode area, nozzle throat diameter.

Starting from Ikarus mission requirements (payload mass, Δv budget, mission duration), derive:

- Required thrust F and specific impulse I_{sp} for each mission phase;
- Propellant mass fraction from Tsiolkovsky equation: $MR = e^{-\frac{\Delta v}{I_{sp} g_0}}$
- Water mass fraction: accounts for three tanks (seawater feed, distilled H_2O , LH_2/LO_2);
- Electrolysis power budget: minimum PV area for closing the propellant production loop in Earth orbit ($r = 1$ AU), Mars orbit ($r = 1.52$ AU) and Saturn orbit ($r = 9.59$ AU).

4.2.2 Thermodynamic and electromagnetic modelling

- Flash distillation: flash fraction ψ vs. feed temperature and chamber pressure [1];
- Electrolysis: Faradaic efficiency η_F and voltage efficiency η_V [2];
- Chemical propulsion: I_{sp} from NASA CEA for H_2/O_2 at $p_c = 2$ –10 bar [7];
- Microwave plasma thruster: thrust and specific impulse as functions of microwave power P_μ and air flow rate $\dot{V}$ after [6]: $F_{ion} = \dot{m}_{ion} v_{ex}$, $v_{ex} = F_{net}/\dot{m}_{air}$

4.2.3 Benchmark comparison (at present)

System	Type	I_{sp} [s]	Propellant	TRL
RL10A-4-2 (Pratt & Whitney) [11]	Expander cycle H_2/O_2	451.0	LH_2/LOX	9
NSTAR (NASA) [12]	Gridded ion	1950–3280	Xenon	9
SABRE (Reaction Engines) [13]	Combined cycle	350–4000	Air/ LH_2	4–5
ATREX (JAXA) [14]	Air-turbo expander	>9000	Air/ LH_2	4–5
ESA ABEP [15]	Air-breathing ion	>1000	Atmosphere	3–4
Skyraker Mode 1 (XSLR)	Expander/aerospike H_2/O_2	>450	ISRU water	2
Skyraker Mode 2 (ICER)	μ -wave ABEP/ N_2/H_2 thruster	3000–10000	ISRU gases	2

Table 5: Skyraker benchmark against flight-proven and research systems.

4.3 Deliverables

- Numerical model in Python & detailed Skyraker system architecture diagram (Fig. 1);
- Feasibility report: for all four subsystems, within the budget and timeline.

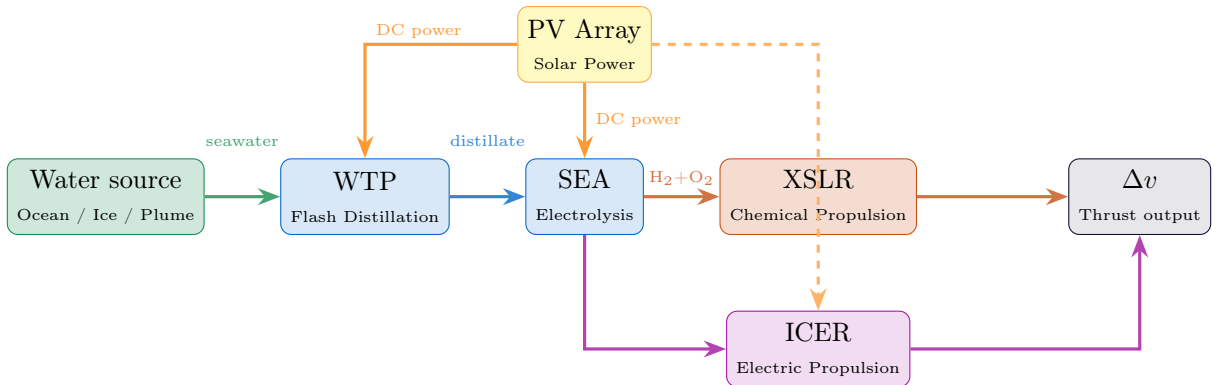


Figure 1: Skyraker system architecture: ISRU chain from water source to thrust output.

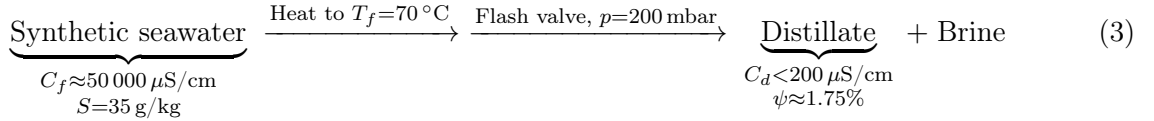
5 Phase 1 – Water Treatment Plant (WTP)

Duration: Months 1–3 **Status:** TRL 3→TRL 5 **Reference:** [1].

5.1 Objectives

Demonstrate that raw seawater-equivalent brine can be processed into distilled water meeting the PEM electrolysis feed specification ($C_d < 200 \mu\text{S}/\text{cm}$, § 1) using a marine-derived flash distillation process. Measure flash fraction ψ , salt rejection R , and specific energy consumption (SEC) experimentally and validate against the thermodynamic model of [1].

5.2 Process Chain



5.3 Key Equations (from dedicated report)

Flash fraction (isenthalpic energy balance):

$$\psi = \frac{c_{p,L}(T_f - T_{\text{flash}})}{\Delta h_{\text{vap}}(T_{\text{flash}})} \approx 1.75\% \quad \text{at the reference operating point} \quad (4)$$

Salt rejection:

$$R = 1 - \frac{C_d}{C_f} > 0.99 \quad (\text{target}) \quad (5)$$

5.4 Hardware

- Borosilicate round-bottom flask 500 mL (boiler)
- Borosilicate Erlenmeyer 500 mL (flash chamber)
- Liebig condenser 300 mm
- Hand vacuum pump + gauge (0–1 bar)
- Ceramic hot plate + magnetic stirrer
- K-type thermocouple + display
- TDS/conductivity pen (C_f , C_d measurement)
- PTFE tape + silicone stoppers + 6 mm silicone tubing

5.5 Deliverables

- Measured ψ_{meas} vs. ψ_{theor} comparison table;
- Salt rejection R at reference operating point;
- SEC curve vs. flash pressure;
- Interface specification: distillate quality → Phase 2 feed.

Phase 1 completion criterion: $C_d < 200 \mu\text{S}/\text{cm}$ demonstrated for at least 3 operating points; $R > 0.95$ at all points; ψ_{meas} within 20% of ψ_{theor} .

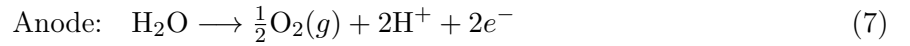
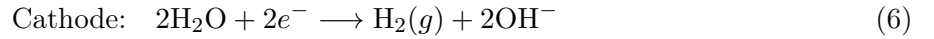
6 Phase 2 – Solar-Electric Assembly (SEA)

Duration: Months 2–4 **Status:** TRL 3→TRL 4 **Reference:** [2]

6.1 Objectives

Demonstrate the electrochemical conversion of distilled water into H_2 and O_2 powered by a photovoltaic source. Measure Faradaic efficiency η_F , voltage efficiency η_V , and gas production rates. Validate against theoretical predictions from Faraday’s laws and the Butler–Volmer model.

6.2 Electrochemical Chain



Gas production rate (from Faraday’s first law):

$$\dot{V}_{\text{H}_2} = \frac{IRT}{2Fp}, \quad \dot{V}_{\text{O}_2} = \frac{1}{2}\dot{V}_{\text{H}_2} \quad (8)$$

6.3 Solar irradiance simulation

The PV source power is varied to simulate solar flux at different heliocentric distances. This produces the $\eta_F(P_{\text{PV}})$ operating envelope relevant to all Space Framework™ mission profiles (see Tab. 6).

Location	r [AU]	Φ [W/m ²]	P_{PV} (relative)
Earth orbit (LEO/GEO)	1.00	1361	1.00
Mars orbit	1.52	589	0.43
Asteroid belt (mean)	2.70	187	0.14
Jupiter orbit	5.20	51	0.037

Table 6: Solar flux and relative PV power at mission-relevant distances.

6.4 Hardware

- 36× AA batteries (4S×9P), ≈ 5.8 V, 26 Ah;
- Faber-Castell TK 9071 2 mm B graphite electrodes;
- Na_2SO_4 anhydrous 0.5 M solution in IKEA PRUTA 1.8 L;
- ONFARMA 60 mL syringes ×2 (gas collection);
- Bromothymol blue (BTB) indicator + pH strips;
- Optional: 10 Ω / 5 W safety resistor, ammeter, voltmeter.

6.5 Deliverables

- η_F vs. time curve (dissolution transient analysis);
- $\dot{V}_{\text{H}_2}$ vs. P_{PV} operating envelope & O_2 production rate confirming 2:1 stoichiometric ratio;;
- pH gradient visualisation (BTB indicator);
- Energy consumption: SEC [Wh/g H_2] compared with commercial AWE/PEM.

Phase 2 completion criterion: 2:1 H_2 : O_2 volume ratio confirmed; $\eta_F > 0.85$ at steady state; $\dot{V}_{\text{H}_2}$ within 15% of Faraday prediction.

7 Phase 3 – Chemical Propulsion Subsystem (XSLR)

Duration: Months 4–7 **Status:** TRL 2→TRL 4 **Reference:** [3]

7.1 Overview

Phase 3 characterises the chemical thruster subsystem in two sequential steps. Phase 3A uses compressed nitrogen as a propellant proxy (cold flow test) to validate the nozzle geometry and thrust measurement system. Phase 3B replaces N_2 with steam generated from WTP subsystem, producing a measurable I_{sp} and closing the conceptual loop from Phase 1–2 to propulsion.

Safety boundary: H_2/O_2 combustion is **not** tested in this phase, at least at first glance. The reasons are: budget constraints, laboratory H_2 safety protocols, and the fact that the LH_2/LOX I_{sp} is a well-validated numerical result (NASA CEA). The gap between steam I_{sp} and LH_2/LOX I_{sp} is quantified analytically and constitutes the *Phase 3 extrapolation* argument.

7.2 Phase 3A – XSLR Cold Flow Test

7.2.1 Objective

Experimentally validate the nozzle geometry (spike/aerospike profile) and the thrust measurement system using compressed nitrogen as an inert propellant proxy. Derive discharge coefficient C_d and thrust coefficient C_F . Experimentally prove the feasibility of the whole system.

7.2.2 Nozzle geometry

The aerospike nozzle is characterised by its altitude-compensating property: the ambient pressure acts on the open base of the spike, maintaining near-ideal thrust across a wide pressure range:

$$C_F = \frac{F}{p_c A_t} \quad F = \dot{m} v_e + (p_e - p_a) A_e \quad (9)$$

where v_e is exit velocity, p_e is exit pressure, p_a is ambient, and A_e is exit area. For an ideal aerospike, the external flow adjusts the effective A_e so that $p_e \approx p_a$ at all altitudes.

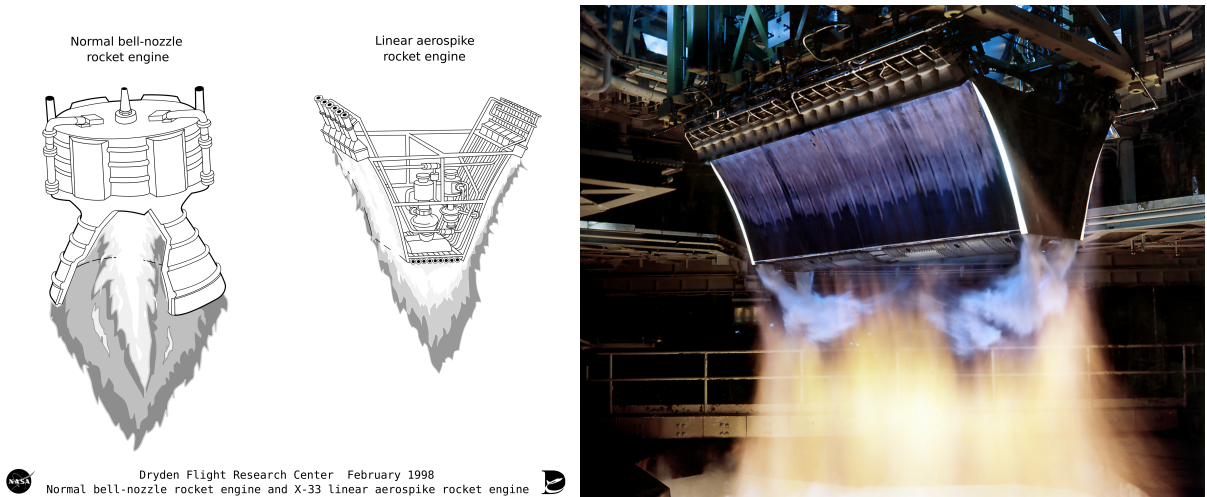


Figure 2: Comparison between the design of a bell-nozzle rocket and an aerospike rocket [17], and XRS-2200 linear aerospike engine for the X-33 program being tested [16].

7.2.3 Hardware

- Spike nozzle: 3D-printed in high-temperature resin or from fiberglass epoxy layup;
- Compressed N₂ cylinder + pressure regulator + mass flow controller;
- Load cell (capacity ≈ 100 g, resolution 0.1 g);
- Lever arm + rigid mount for thrust measurement;
- Arduino board for data acquisition;
- Pressure transducer (0–10 bar) for chamber pressure.

7.2.4 Deliverables

- Thrust F vs. chamber pressure p_c curve;
- Discharge coefficient C_d vs. p_c ;
- Comparison with CEA/analytical model;
- Nozzle geometry validation.

7.3 Phase 3B – XSLR Steam Flow Test

7.3.1 Objective

Demonstrate water-fed propulsion using steam as a proxy for the full expander cycle. Measure effective I_{sp} and compare with CEA predictions for both H₂O and H₂/O₂ propellants.

7.3.2 Experimental setup

Replace the N₂ supply with a resistively-heated steam generator:

$$P_{\text{heater}} = \dot{m}_w \cdot [c_p(T_{\text{sat}} - T_{\text{in}}) + \Delta h_{\text{vap}}(T_{\text{sat}})] \quad (10)$$

The steam enters the aerospike chamber at p_c and expands through the nozzle. Then:

$$I_{sp} = \frac{F}{\dot{m}_{\text{prop}} g_0} \quad (11)$$

Expected values (CEA, $p_c = 5$ bar):

- H₂O steam: $I_{sp,\text{steam}} \approx 150\text{--}190$ s;
- H₂/O₂ (theoretical): $I_{sp,\text{LH}_2/\text{LOX}} \approx 420\text{--}465$ s.

The gap $\Delta I_{sp} = I_{sp,\text{LH}_2/\text{LO}_2} - I_{sp,\text{steam}}$ motivates the future transition to cryogenic propellant.

7.3.3 Hardware additions for 3B

- Electric resistance boiler (stainless SS316 chamber, 200–500 W);
- PID temperature controller;
- Check valves + safety relief valve (set at $1.2 p_c$).

7.3.4 Deliverables

- $I_{sp,\text{meas}}$ vs. p_c curve;
- Thrust F vs. steam mass flow rate;
- Gap analysis: $I_{sp,\text{steam}}$ vs. $I_{sp,\text{LH}_2/\text{LOX}}$ (CEA);
- Nozzle thermal characterisation.

Phase 3 completion criterion: $F > 0$ mN measured with load cell; $I_{sp,\text{meas}}$ within 25% of CEA prediction for steam; repeatability $< 10\%$ over 5 successive firings.

8 Phase 4 – Integrated Chemical Demonstration

Duration: Months 7–9 **Target:** TRL 4/4.5 **Reference:** This report

8.1 Objective

Integrate all subsystems demonstrated in Phases 1–3 into a single operating chain and demonstrate the complete Skyraker ISRU propulsion concept:

$$\text{Seawater} \longrightarrow \text{WTP} \longrightarrow \text{SEA} \longrightarrow \text{Hot H}_2\text{O steam flow} \longrightarrow \text{Thrust} \quad (12)$$

Note on TRL: In this demonstration, the thruster uses steam as propellant. Successfully combusting electrolytic H_2 (a **TRL 5** milestone) in the same fiberglass-epoxy chamber at high O/F ratio with $< 10\%$ H_2 dilution in N_2 would place the system towards **TRL 5**.

8.2 Measurements and Data

- M1. End-to-end water balance:** measure $\dot{m}_{\text{feed}}$, $\dot{m}_{\text{distillate}}$, $\dot{m}_{\text{brine}}$ and verify mass conservation.
- M2. Energy audit:** measure P_{PV} , P_{WTP} , P_{SEA} , P_{thruster} and compute system efficiency η_{sys} .
- M3. Coupled distillate quality:** verify that WTP output fed directly to SEA produces $\eta_F > 0.85$.
- M4. Thrust continuity:** for > 60 s continuous operation and restart capability.

8.3 Deliverables

- Complete dataset: all Phase 1–3 measurements repeated in integrated mode;
- System efficiency $\eta_{\text{sys}} = F \cdot v_e / P_{\text{tot}}$ and power budget;
- Discrepancy analysis: integrated vs. standalone subsystem performance;
- Architecture diagram with validated parameters at each interface.

Phase 4 completion criterion: Demonstration of the complete chain for chemical propulsion subsystem only in a single integrated laboratory session, with all metrics within tolerances.

9 Phase 5 – ICER & Full System Integration

Duration: Months 9–12 **Target:** TRL 4–5 **References:** [4], [6].

9.1 Rationale

The ICER subsystem provides Skyraker’s high- I_{sp} electric propulsion (Mode 2a/2b). Phase 5 builds a **microwave air plasma thruster** operating at atmospheric pressure, directly following the design of Ye et al. (2020) [6]. This approach requires no vacuum chamber, making it the only plasma propulsion technology demonstrable in a standard university laboratory at **TRL 4–5**.

9.2 Working Principle (after Ye et al. 2020)

A 2.45 GHz magnetron (salvaged from a domestic microwave oven) concentrates the microwave field in a handmade aluminium waveguide ($90 \times 50 \times 600$ mm, flattened to 25 mm at the ionisation cavity). A borosilicate tube ($\varnothing_i = 24$ mm, $L = 600$ mm) passes vertically through the cavity. Compressed air enters tangentially as a vortex, stabilising the plasma and protecting the tube wall. The microwave field ionises the air, producing a high-temperature plasma jet that generates thrust.

The net propulsion force (subtracting the air-only baseline) and net jet pressure generated by the thruster over the tube cross-section are:

$$F_{\text{net}} = F - F_0 = (M - M_0)g, \quad P_{\text{net}} = \frac{F - F_0}{\pi R^2}, \quad R = 12 \text{ mm} \quad (13)$$

At 400 W and 1.45 m³/h air flow, Ye et al. measured $F_{\text{net}} \approx 11 \text{ N}$ and $F/P \approx 28 \text{ N/kW}$, comparable to conventional jet engines at the same power.

9.3 Effective Specific Impulse

Unlike a gridded electrostatic thruster, the microwave plasma jet accelerates gas thermally. The effective exhaust velocity and specific impulse are:

$$v_{\text{eff}} = \frac{F_{\text{net}}}{\dot{m}_{\text{air}}}, \quad I_{\text{sp, meas}} = \frac{F_{\text{net}}}{\dot{m}_{\text{air}} g_0} \quad (14)$$

Expected range at 400 W: $I_{\text{sp}} \approx 50\text{--}200 \text{ s}$ depending on air flow and power. Significantly lower than a gridded ion thruster but sufficient to demonstrate the ABEP principle.

9.4 Hardware

- Magnetron assembly (2.45 GHz, $\geq 800 \text{ W}$); retain HV transformer, capacitor, fan, interlocks;
- Handmade Al waveguide, flattened cavity, 3-stub tuner, short-circuit end plate;
- Borosilicate tube $\varnothing_i = 24 \text{ mm}$, $L = 600 \text{ mm}$;
- Vortex air inlet (tangential SS316 collar + 6 mm Cu elbow);
- Air compressor ($\geq 30 \text{ L/min}$) + rotameter 0–2 m³/h;
- Steel ball force gauge: hollow steel ball $D > 28 \text{ mm}$, small calibrated steel beads (M3 bolts);
- Microwave leakage detector (2.45 GHz, **mandatory**);
- HV discharge resistor 10 k Ω /10 W (capacitor discharge, **mandatory**);
- Load cell + HX711 + Arduino (reused from Phase 3A) for optional continuous thrust recording.

9.5 ABEP Mode Simulation

To validate the Skyraker Mode 2b (ABEP) principle, throttle air flow to $\dot{V} < 0.3 \text{ m}^3/\text{h}$, simulating a dilute tenuous atmosphere. Record the minimum power for sustained plasma ignition and the $F_{\text{net}}/\dot{m}_{\text{air}}$ figure of merit vs. $P_{\mu}/\dot{m}_{\text{air}}$.

Phase 5 completion criterion (ICER): $F_{\text{net}} > 0$ at ≥ 3 power/flow combinations; F/P within 50% of Ye et al. reference (28 N/kW); ABEP simulation: sustained plasma at $\dot{V} < 0.3 \text{ m}^3/\text{h}$; no microwave leakage above ICNIRP limit.

9.6 Full System Integration

After Phase 5 standalone commissioning, the ICER is integrated into the full demonstrator chain:

$$\text{Seawater} \rightarrow \text{WTP} \rightarrow \text{SEA} \rightarrow \begin{cases} \text{H}_2\text{O} \rightarrow \text{XSLR steam thruster} \\ \text{H}_2 \rightarrow \text{ICER microwave plasma} \end{cases} \rightarrow \Delta v \quad (15)$$

Both thrusters operated simultaneously in a single laboratory session constitutes the **TRL 4** milestone for the full Skyraker system (**TRL 4.5** if also dilute combustion is performed).

System TRL 4/4.5 completion criterion: Demonstration of the complete ISRU-to-thrust chain (Fig. 1) with XSLR and ICER operating within specified tolerances in a single integrated session, with steam and/or dilute combustion.

10 9-Month Schedule

Phase	Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9
0	Architecture definition	–								
1	WTP: Flash distillation demo	–	–	–						
2	SEA: Electrolysis demo		–	–	–					
3A	XSLR Cold Flow Test (N ₂)				–	–				
3B	XSLR Steam Flow Test					–	–	–		
4	Integrated Chemical							–	–	
5	ICER + full system								–	–
W	Report writing (parallel)		–	–	–	–	–	–	–	–

Table 7: 9-month development schedule.

11 Risk Register

Risk	Prob.	Impact	Mitigation
Flash fraction too low to measure accurately	Medium	Medium	Lower p_{flash} to 100 mbar; increase T_f to 80–85 °C
H ₂ dissolution delays η_F measurement	High	Low	Wait 2–3 min for saturation; expect transient in early data
Aerospike print warps under steam temperature	Medium	High	Prototype in PLA first for geometry check
Insufficient load cell resolution	Low	High	Use lever arm amplification; calibrate with known weights
PEM electrolyte contamination from tap water	Low	Medium	Always use demineralised water; weekly conductivity check
H ₂ accumulation in enclosed space	Low	High	Forced ventilation; H ₂ sensor; never operate alone (two-person rule)
Budget overrun in Phase 3B	Medium	Medium	Source N ₂ cylinder via university lab
Ion thruster HV safety	Medium	High	University HV safety protocol; interlocks; discharge capacitors before handling

Table 8: Skyraker development risk register.

12 Future work

The following developments are explicitly scoped out of the present roadmap and constitute the natural evolution towards **TRL 5–TRL 6**:

- **LH₂/LOX thruster** – Replace the steam flow with pressure-rated hardware and dedicated safety facilities for combustion between the electrolytic H₂ and O₂.
- **Aerospike optimisation** – Full aerodynamic optimisation of the spike contour using CFD. Altitude-compensation characterisation in simulation chamber.

- **Multi-stage flash WTP** – Three-stage flash system with brine heat recovery ($\text{GOR} > 4$). Scales up water production rate for propellant generation.
- **Ion thruster in vacuum** – Phase 5 demonstrator test in vacuum ($p < 10^{-4}$ mbar).
- **Integrated mission simulation** – Full Python model with trajectory integration.
- **Battery pack integration** – Battery pack as auxiliary power source, to be used in shadowed regions during spaceflight. No new experimental work at this stage since past missions heritage.
- **Ikarus structural prototype** – First structural mock-up of Ikarus airframe using composite.

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